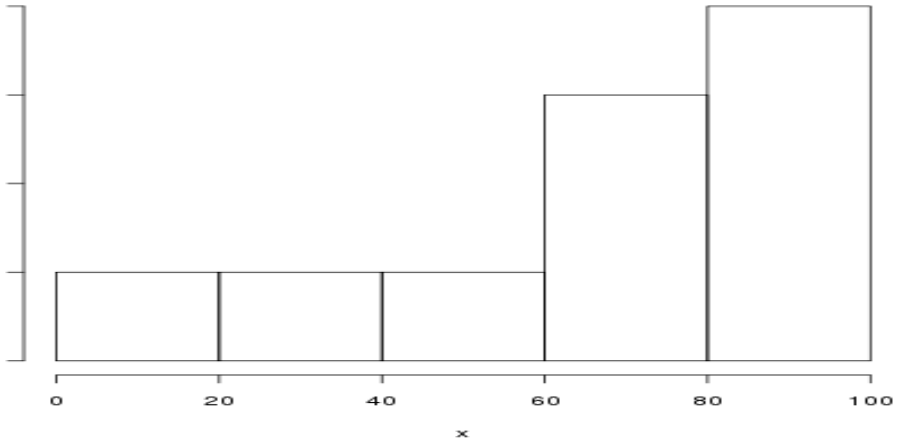
Review Questions for Lecture 4

1. Suppose you are given a vector in R named "coffee" that contains data on how many coffees each student in Stat 280 drinks per week. What R code would you write if you wanted to create a proper histogram (using a density scale) of this data and you wanted to have one bar centered over every integer? Assume that the maximum number of coffees anyone drinks per week is 50.

2. Which three percentiles do you learn from a boxplot?

3. Is it possible for a boxplot to appear without any whiskers? If so, provide an example of a list of numbers that would produce such a boxplot. If not, explain why not.

Questions 4-11 refer to the histogram below. Assume that the data is spread evenly within each block.



4. True or false and explain: "The 60th percentile is twice the 30th percentile."
5. True or false and explain: "The 30th percentile is twice the 15th percentile."
6. True or false and explain: "The median is less than the mean."
7. True or false and explain: "The mean is greater than 50."
8. True or false and explain: "The standard deviation is greater than the mean."
9. What is the median? Please give an answer that is correct to at least one decimal place.
10. What is the interquartile range? Please give an answer that is correct to at least one decimal place.
11. If a histogram follows the normal curve, what is greater, the SD or the IQR? Explain.

Note that Questions 12 – 18 use material from Lecture 5

12. The scale on a midterm is from 0 to 100. The average score is 80. What is the largest possible fraction of the class that got higher than 95?
13. (Tricky!) A midterm is graded on a scale of 0 to 100. The average score is 80. What is the largest possible fraction of the class that got lower than 65?
14. The scale on a midterm is from 0 to 100. The average score is 80. The standard deviation is 5. What is the largest possible fraction of the class that got higher than 95 or less than 65?
15. Give an example of a list where the IQR is greater than the SD and where the SD is greater than the IQR.
16. Suppose the average house price in a certain city is $200,000. What is the largest possible fraction of houses that cost more than $1,000,000? Is your answer realistic? If not, is it too big or too small? Why?
17. Suppose that *X* is a list of non-negative numbers. Use the techniques discussed in lecture to show that the median of *X* is at most twice the mean of *X*.
18. Suppose that *X* is a list of numbers with standard deviation *s*. Use the techniques discussed in lecture to show that $|median(X) - \overline{X}| \leq s\sqrt{2}$. Note that $\overline{X}$ refers to the mean of *X*. One can use other techniques to show that $\left|median(X) - \overline{X}\right| \leq s.$.